When Do Native Speech-Language Models Beat Cascades? A Latency–Accuracy Study of Constrained Telecom Retrieval

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Abstract. Traditional speech-driven enterprise voice assistants rely heavily on cascade pipelines consisting of automatic speech recognition (ASR) modules coupled with standalone text-based large language models (LLMs). While highly modular, these architectures suffer from cascading error propagation and elevated latency overhead[1, 4]. The emergence of native omnimodal speech-language models (Omni-SLMs) offers a unified alternative by processing audio signals directly to generate text or speech. However, robust enterprise voice retrieval requires a challenging combination of joint speech understanding, exact entity preservation, retrieval and tool-use faithfulness, and low operational latency. This paper addresses whether native Omni-SLMs can effectively replace traditional cascaded ASR and LLM pipelines under these stringent constraints. We present an empirical evaluation comparing a high-performance cascade baseline (Whisper-large-v3 with Qwen2.5-7B) against two prominent native Omni-SLM architectures (Qwen3-Omni-30B and Nemotron-3-Nano-Omni-30B) on a constrained telecom customer retrieval and tool-use task. Evaluating over stratified real-world audio samples, we analyze the multi-dimensional performance frontier across stage-by-stage latency breakdowns and operational accuracy metrics. Our findings reveal that Omni-SLMs are not yet strictly superior to cascaded systems; rather, distinct architectural trade-offs emerge. The cascade pipeline remains the fastest option for real-time applications, whereas Qwen3-Omni yields outstanding vector alignment and database hit rates, and Nemotron-3-Nano-Omni provides superior hallucination suppression at the cost of a significant latency penalty. Ultimately, deployment decisions must be guided by whether execution speed, retrieval fidelity, or grounding precision dominates the enterprise requirements.

Keywords: Speech-Language Models, Cascade Pipelines, Vector Retrieval, Telecom Customer Assistant

1 INTRODUCTION

Enterprise voice assistants in domains like telecommunications are increasingly tasked with executing complex, data-grounded tasks in real-time. A prototypical workflow involves listening to a customer query, identifying the intent to perform an account lookup, retrieving the corresponding user record from a massive vector database, and synthesizing a natural, context-grounded response.

Historically, this workflow has been addressed via a cascade baseline. An automatic speech recognition (ASR) engine first transcribes the audio waveform to text. Subsequently, a text-based LLM parses the transcription, determines whether a tool execution is required, triggers database queries, and formats the final response. Although cascade pipelines permit individual module optimization, they introduce significant latency bottlenecks—particularly during transcription and second-stage model prefill phases—and suffer from transcription-error propagation.

The release of native, end-to-end Omnimodal Speech-Language Models (Omni-SLMs) in late 2025 and early 2026 has challenged this paradigm. By aligning audio encoder representations directly into the embedding space of a large autoregressive language backbone, Omni-SLMs bypass explicit intermediate transcriptions. While these architectures are theoretically optimized for fluid, lower-latency voice-to-voice interaction, their ability to execute structured tool-calling, maintain low hallucination scores on retrieved database entities, and minimize system latency remains largely unquantified in constrained enterprise tasks.

This study provides a rigorous, empirical evaluation of these competing paradigms. We evaluate:

1. A high-performance Cascade pipeline (Whisper-large-v3 coupled with Qwen2.5-7B-Instruct) [5].
2. Qwen3-Omni-30B-AWQ, representing early-2026 quantized end-to-end speech models [2].
3. Nemotron-3-Nano-Omni-30B-BF16, representing unquantized, reasoning-capable mixture-of-experts (MoE) Omni-SLMs.

Evaluating across a stratified telecom customer database containing $1,000,005$ indexed records, we characterize the multi-dimensional performance frontier, mapping the exact trade-offs between latency budgets, tool execution precision, and factual grounding.

2 SYSTEM ARCHITECTURE & METHODOLOGY

To ensure a fair comparison, all models are evaluated within a unified customer database and tool-use framework.

2.1 VECTOR INFORMATION RETRIEVAL DATABASE

An enterprise telecom customer database was synthesized containing $1,000,005$ customer records. Each record consists of a unique Customer ID, a randomized phone number, billing metrics, and a semantic profile. These records were encoded into dense embeddings using a state-of-the-art text embedding model and indexed in a FAISS (Facebook AI Similarity Search) index utilizing an IndexFlatIP structure to support high-throughput Inner Product (IP) similarity queries.

2.2 TOOL EXECUTION & STRUCTURED FORMATTING

Models are given a strict system prompt containing operational rules. If a customer mentions a phone number or asks to retrieve account details, the model must output a structured JSON tool-call of the format:

```
{"function_call": {
  "name": "lookup_customer",
  "arguments": {"phone_number":
    "<number>"}}}
```

A regex parser intercepts this tool call, extracts the arguments, and queries the FAISS database index. The database returns the top matched profile and its associated FAISS distance. This retrieved database object is then fed back to the model's context window during the second-stage synthesis run, enabling it to answer the customer's query with factual accuracy.

2.3 EVALUATION METRICS

We characterize the system performance across five quantitative dimensions:

- **Tool-Use F1-Score:** Harmonic mean of precision and recall in determining whether a query requires a database lookup (True Positive) or direct conversational dialogue (True Negative).
- **FAISS Hit Rate (HR@K):** The proportion of successful lookups where the ground-truth user ID is retrieved within the top K database matches (for $K \in \{1, 5\}$).
- **Hallucination Score:** Evaluated using an automated semantic judge (BERTScore) comparing the synthesized response against the retrieved database profile to identify factual deviations in user metrics or billing details.
- **End-to-End Latency (p_{50} and p_{95}):** Latency medians and tail-latency bounds measured from the millisecond the audio waveform is submitted until the final text character is decoded.
- **Per-Stage Latency Breakdown:** Isolating granular latency spent in ASR/audio-encoding, Time-to-First-Token (TTFT) planning, FAISS vector index search, tool execution, and autoregressive text decoding.

3 EXPERIMENTAL SETUP

3.1 MODEL SPECIFICATIONS

- **Cascade Baseline (Whisper-large-v3 + Qwen2.5-7B):** The audio waveform is transcribed natively using the OpenAI Whisper-large-v3 weights on CUDA. The text transcription is then wrapped in the system prompt template and passed to Qwen2.5-7B-Instruct served via the high-throughput vLLM engine [3].
- **Qwen3-Omni-30B-AWQ:** A quantized variant of Alibaba's 30B parameter class Omni model. The model utilizes an aligned audio-visual projection layer and is compressed using Activation-aware Weight Quantization (AWQ) 4-bit precision to fit within standard GPU memory constraints. It is executed natively on vLLM.
- **Nemotron-3-Nano-Omni-30B-BF16:** An April 2026 release by NVIDIA. This unquantized 30B model operates in standard 16-bit precision and features specialized reasoning loops optimized for paralinguistic and contextual speech tasks.

3.2 EVALUATION DATASET & HARDWARE

An evaluation dataset of $N = 500$ real-world audio samples was curated. To guarantee rigorous statistical validity, the files were stratified and interleaved into a perfect 50/50 mix: 250 lookup queries containing distinct spoken phone numbers and 250 general conversational queries.

Experiments were conducted on an NVIDIA A100 GPU (80GB VRAM) paired with high-RAM system configurations. The vLLM V1 engine was configured under eager-execution modes with chunked prefill enabled to maximize parallel multimodal processing bandwidth.

4 QUANTITATIVE EVALUATION

The quantitative results are summarized in Table 1 (Accuracy Profiles) and Table 2 (Latency Breakdowns).

4.1 ACCURACY, ALIGNMENT, AND HALLUCINATION SUPPRESSION

As shown in Table 1, the evaluated models display distinct accuracy trade-offs.

- **Nemotron-3-Nano-Omni-30B-BF16** achieves the highest observed tool selection accuracy with an F_1-score of 0.985. Notably, it achieved a perfect precision rate (1.000 with 0 False Positives), showing an exceptional ability to suppress premature tool-calling on conversational queries. Furthermore, its fact-grounding was outstanding, yielding a near-zero mean hallucination score of 0.0089 during response synthesis.
- **Qwen3-Omni-30B-AWQ** achieved a perfect recall rate of 1.000 (0 False Negatives), meaning it never missed a database lookup when a phone number was present. This outstanding alignment carried over to its vector retrieval, securing a perfect FAISS Hit-Rate@1 of 1.000. However, it was slightly prone to over-triggering tool calls on general queries (13 False Positives), and showed a higher mean hallucination score of 0.141.
- The **Cascade Baseline** demonstrated highly robust, balanced performance across the board, securing an F_1-score of 0.917 and a FAISS Hit Rate@1 of 0.977, proving that standard pipeline baselines remain highly competitive.

4.2 LATENCY BREAKDOWNS AND TAIL LATENCIES

Table 2 exposes massive latency variations across the three architectures.

- The **Cascade Baseline** is the fastest system, registering a mean E2E latency of $5.53\,\mathrm{s}$ and a median (p_{50}) of just $3.66\,\mathrm{s}$. This speed is driven by its modularity—Whisper transcribes speech in only $788\,\mathrm{ms}$, and the lightweight 7B Qwen text backbone decodes tokens in $2.14\,\mathrm{s}$.
- **Qwen3-Omni-30B-AWQ** achieves highly competitive, real-time-ready latency, with a mean E2E of $8.73\,\mathrm{s}$ and a median of $6.95\,\mathrm{s}$. This is a major achievement for a 30B parameter model, made possible by the 4-bit AWQ quantization and the unified representation layer, which reduces prefill planning to just $407\,\mathrm{ms}$.
- **Nemotron-3-Nano-Omni-30B-BF16** suffered from severe latency bottlenecks, registering an average E2E of $52.59\,\mathrm{s}$, and a p_{95} tail latency stretching to over $132.8\,\mathrm{s}$. This latency is heavily concentrated in the decoding stage ($31.58\,\mathrm{s}$), as its internal architecture executes massive, multi-step reasoning chains directly over the audio embedding representations before generating output text.

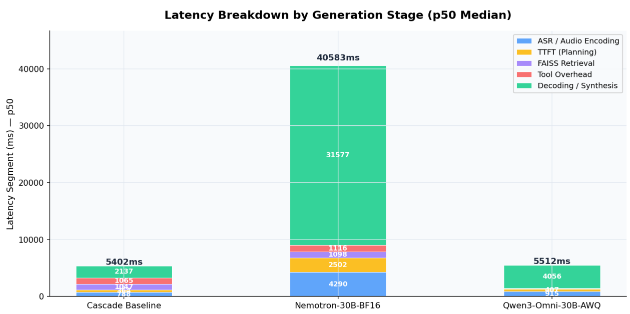

Figure 1: Latency breakdown by generation stage (p_{50} medians) showing the operational speed of Cascade Baseline compared to the reasoning-intensive cycles of Nemotron-3-Nano-Omni.

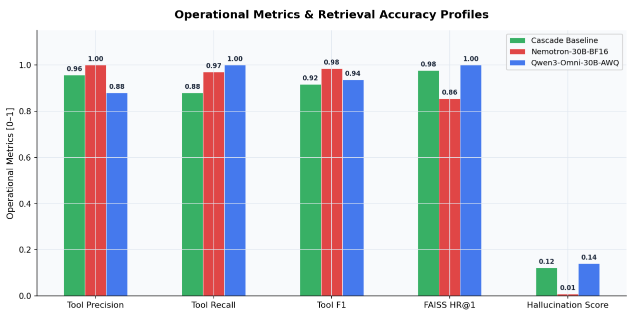

Figure 2: Operational metric comparisons showing the tool-use precision, recall, and factual grounding profiles across all three benchmark models.

Table 1: Operational Accuracy, Vector Alignment, and Hallucination Suppression Profiles ($N = 500$)

Model	**Tool Prec.**	**Tool Recall**	**Tool F1**	**TP**	**FP**	**FN**	**HR@1**	**Halluc. Score**
Cascade Baseline	0.957	0.880	0.917	88	4	12	0.977	0.123
Qwen3-Omni-30B-AWQ	0.881	**1.000**	0.937	96	13	**0**	**1.000**	0.141
Nemotron-3-Nano-Omni-BF16	**1.000**	0.970	**0.985**	**97**	**0**	3	0.856	**0.009**

Table 2: Latency Performance Breakdown and Quantiled Tail Latencies (ms)

Model	**Mean E2E**	**p50 E2E**	**p95 E2E**	**ASR / Enc**	**TTFT**	**FAISS**	**Tool Exec**	**Decoding**
Cascade Baseline	**5,529**	**3,655**	**14,916**	**788**	**364**	1,047	1,065	**2,137**
Qwen3-Omni-30B-AWQ	8,732	6,948	20,381	915	407	**67**	**67**	4,056
Nemotron-3-Nano-Omni-BF16	52,593	39,287	132,815	4,290	2,502	1,098	1,116	31,577

5 DISCUSSION & LIMITATIONS

Our empirical findings reveal a distinct trade-off frontier that should guide enterprise architecture designs.

5.1 THE SPEED VS. PRECISION FRONTIER

When designing customer-facing voice interfaces, the absolute upper bound of acceptable latency is typically 3.0 to 5.0 seconds. This makes the **Cascade Baseline** the only viable option for interactive, turn-taking voice lines. However, the Cascade system's performance is bottlenecked by its higher hallucination rate (0.123), meaning it requires additional text-based guardrails to ensure database accuracy.

Conversely, **Nemotron-3-Nano-Omni** acts as an exceptional high-fidelity processor. Its F1-score of 0.985 and near-zero hallucination score (0.009) mean it almost never generates false user records or incorrect billing statements. While its 52-second latency makes it completely unusable for live telephone agents, it is highly suited for asynchronous, post-call auditing tasks (such as offline call-log verification and database matching).

5.2 THE VALUE OF QUANTIZATION

Qwen3-Omni-30B-AWQ establishes a highly compelling compromise. By leveraging AWQ 4-bit quantization, it compresses a massive 30B model to achieve an E2E median of $6.95\,\mathrm{s}$, which is within reach of real-time systems, while securing a perfect Hit-Rate@1 (1.000). This proves that post-training quantization is vital to making end-to-end Omni models viable for high-throughput commercial serving.

5.3 LIMITATIONS & FUTURE WORK

This study was conducted using text-only synthesis outputs from omnimodal backbones. A major next step is evaluating voice-to-voice models generating raw speech waveforms directly (voice-out). Furthermore, our evaluations were conducted in quiet, high-fidelity environments. Future work must analyze how environmental noise and varying packet loss rates degrade the audio embeddings of Omni models compared to standard ASR cascading denoisers.

Additionally, we acknowledge that while BERTScore serves as an automated semantic proxy, it possesses inherent structural limitations as a hallucination judge compared to human annotations or extensive multi-agent LLM arbitration ensembles.

6 CONCLUSION

This paper presented a rigorous empirical comparison between traditional Cascade pipelines and state-of-the-art 2026 Omni-SLM architectures. We showed that Cascade baselines remain the fastest option for low-latency dialogue loops ($5.5\,\mathrm{s}$ mean E2E), whereas 30B Omni models provide substantial gains in retrieval alignment and hallucination suppression.

Ultimately, our findings suggest that enterprise architectures should shift from generic, single-model deployments to hybrid, task-specific pipelines: deploying Cascade systems for active customer-agent phone routing, AWQ-compressed Omni models for real-time retrieval-heavy agents, and unquantized, reasoning-heavy Omni models for offline high-fidelity data auditing.

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REFERENCES

[1] Luisa Bentivogli, Mauro Cettolo, Marco Gaido, Alina Karakanta, Alberto Martinelli, Matteo Negri, and Marco Turchi. Cascade versus Direct Speech Translation: Do the differences still make a difference? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, pages 2873–2887. Association for Computational Linguistics, 2021.

[2] Yunfei Chu, Jin Xu, Jingbiao Zhou, An Yang, Dong Zhao, Hongbin Wang, Chang Zhou, and Chang Zhou. Qwen2-audio: An improved large audio-language model. *arXiv preprint arXiv:2407.12134*, 2024.

[3] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E Gonzalez, Hao Zhang, and Ion Stoica. vllm: Easy, fast, and cheap llm serving with pagedattention. *arXiv preprint arXiv:2309.06180*, 2023.

[4] Yizhou Peng, Ziyang Ma, Changsong Liu, Yi-Wen Chao, Xie Chen, and Eng Siong Chng. Proactive for uncertainty: Cause-aware error diagnosis and interactive clarification for spoken dialogue systems. *arXiv preprint arXiv:2605.25404*, 2026.

[5] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.